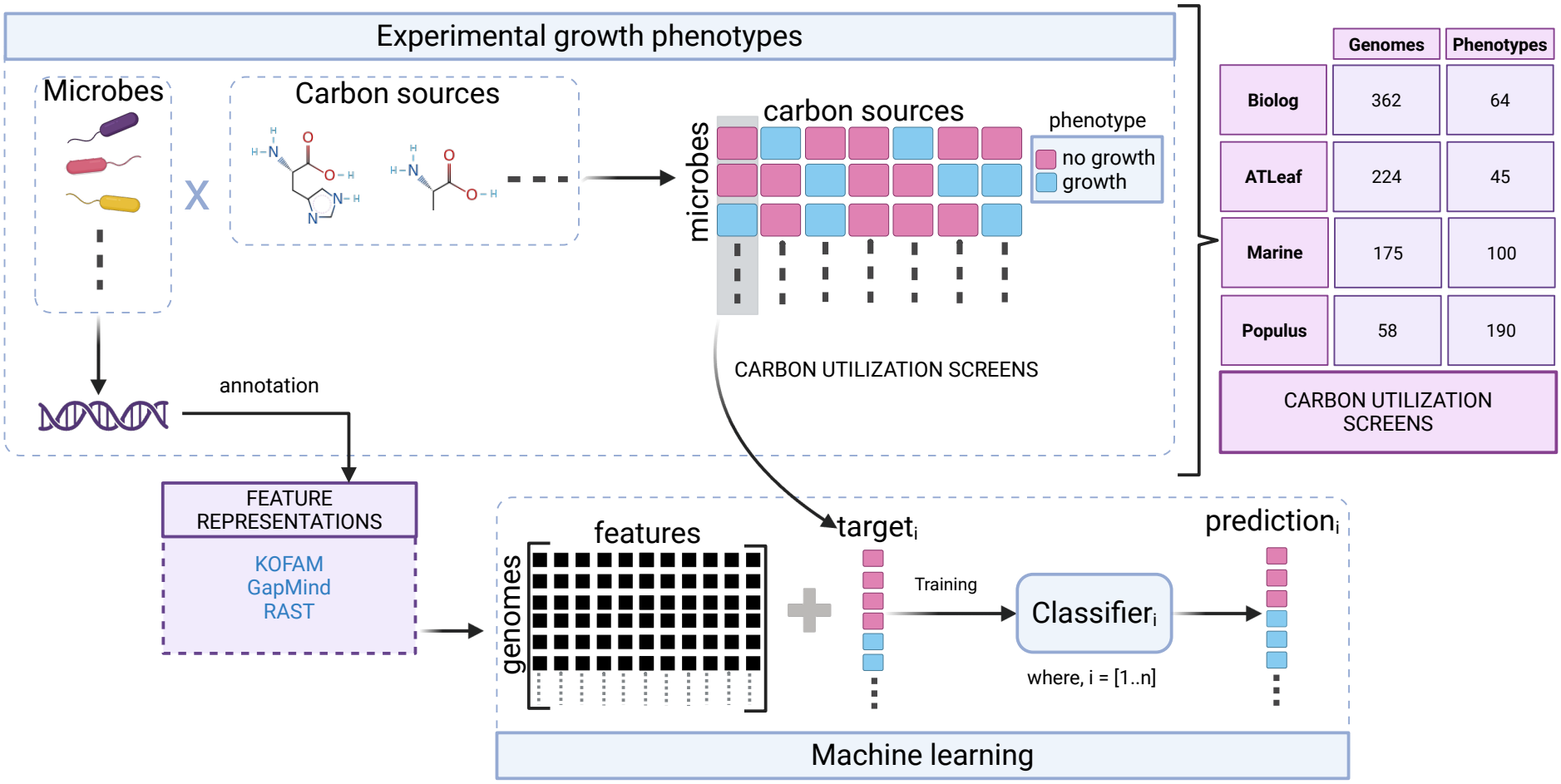


Created in BioRender. Kishore, D. (2026) https://BioRender.com/tz7t8n7

Approach	Training data	Features	Evaluation	Balanced accuracy	Finding
Random holdout	full	KOFAM	in-distribution	med 0.86, IQR 0.81-0.87	Optimistic; inflated by related genomes in test
Cross-dataset (full data)	full	KOFAM	cross-dataset	med 0.62, IQR 0.60-0.66	Overfits; transferable features for only 6/15
Concordance filtering	concordant	KOFAM	cross-dataset	med 0.68, IQR 0.61-0.77	**Recovers pathway genes; transferable for 12/15**
Feature combination	full	**combined (~17k)**	cross-dataset	med 0.66, IQR 0.60-0.69	Favours curated (7.6×), yet loses to curated alone
Feature filtering	full	**filtered GapMind (~ 34)**	cross-dataset	med 0.70, IQR 0.64-0.77	Highest median, but capped by the curated pathway

